MODEL FAIRNESS PASSPORT · AUDIT BEYOND ACCURACY

Brain-age model SFCN · IXI

An independent audit of accuracy, group fairness, acquisition robustness, explanations and uncertainty.

Passport no. **BAF-XAI-2026-001**
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Checkpoint **sha256 13ddd261…**
Valid for **IXI-like T1 MRI**

ARCHITECTURE	PRETRAINING	FINE-TUNED ON	EVALUATED ON
SFCN, 3D CNN 72 age bins (18–90)	**UK Biobank** Peng et al. 2021	**IXI, 339 adults** 3 London sites	**73 test + 73 val** test used once

PASS Accuracy

PASS Fairness by sex

FAIL Fairness by site

FAIL Scanner invariance

PASS Faithful explanations

PASS Honest uncertainty

PERFORMANCE & FAIRNESS

CHECK	RESULT	95% CI
● Test MAE	3.71 yrs · r 0.966	3.10 – 4.38
● Gap, male vs female	−0.59 yrs	−2.67 – +1.64
● Gap, HH vs Guys	+1.06 yrs	−1.44 – +3.53
● Gap, IOP vs Guys	**−5.06 yrs**	−8.66 – −1.34
● Error, any group	no difference	all CIs span 0
● IOP unseen (extra error)	**+3.30 yrs**	+1.62 – +4.93
● HH unseen (extra error)	+0.25 yrs	−0.88 – +1.36

ROBUSTNESS, EXPLANATIONS, UNCERTAINTY

CHECK	RESULT
● Site read from features	**91%** (chance 33%)
● Scan look alone shifts age	**3–4 yrs**, 100% same direction
● IOP offset explained by intensity / blur / fingerprint	no: ≤1 / 0 / persists
● Sanity check (randomised layers)	ρ 0.13, maps destroyed
● Deletion test, top vs random 5%	4.83 vs 1.04 yrs
● Uncertainty on unseen IOP	doubles, 92% coverage

USAGE GUIDANCE

SUITABLE FOR

- Research on brain ageing within a single acquisition site
- Comparing groups scanned on the same scanner and protocol
- Teaching and methods development

NOT SUITABLE FOR

- Any clinical or individual decision
- Comparing brain-age gaps *across* scanners without site-aware validation
- New sites without checking its uncertainty first

KNOWN LIMITATIONS

- Small subgroups (9 IOP test subjects); single seed and split. The IOP offset nonetheless replicates in validation (−4.1) and test (−5.1).
- In IXI, site is confounded with who each site recruited: the cause of the IOP offset (untested scanner properties or cohort) cannot be separated.
- Healthy volunteers aged 20–86 only. Attribution resolution 8 mm. Uncertainty is the model's own spread, not a calibrated interval.

Evidence: every figure on this page comes from the scripts in **scripts/03b–14** and is logged with dates in **PROJECT_GUIDE.md**. Statistics: age-adjusted OLS with HC3 errors, 5,000-sample bootstrap CIs, FDR where several regions are tested. Wording: acquisition-site effects, not "hospital bias"; no clinical claims.

AUDITED beyond accuracy